

Lecture (9): Nitrogen

Nitrogen (Dinitrogen) N_2 is a colourless and odourless gas (condensation at 77K) Dinitrogen makes up 78% of the Earth's atmosphere, and can be obtained by liquefaction and fractional distillation of air, It has two isotopes in nature, N^{15} and N^{14} .

Nitrogen is relatively unreactive because the bond enthalpy of the nitrogen–nitrogen triple bond is very high (944 kJ/mol). $N_{2(g)}$ is usually used as an inert atmosphere for reactions that cannot be carried out in oxygen.

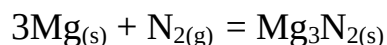
Many important gases cannot be liquefied at room temperature. Nitrogen, for example, has a critical temperature of 147°C . This means the gas cannot be liquefied until the temperature is below 147°C .

Properties and Uses of Nitrogen Daniel Rutherford, a chemist and physician, discovered nitrogen, N_2 , in air in 1772. In his experiments, he removed the oxygen from air by burning a substance in it. When burning carbon-containing substances, he removed the carbon dioxide that formed by reacting it with aqueous potassium hydroxide. He then showed that the residual gas would no longer support either combustion or living organisms. Although this residual gas is mostly N_2 it does contain small amounts of noble gases.

Liquid nitrogen (bp 77 K) is frequently used as a coolant in the laboratory or in industry. Also can be used for the quick freezing of foods. The element nitrogen is crucial to life: it is a component of all proteins, which are involved in almost every biochemical process that occurs in living organisms.

Reaction of Nitrogen:

Some very reactive metals do react directly with nitrogen, however. For example, when magnesium metal burns in air, it forms the nitride, as well as the oxide



One of the rare reactions of nitrogen at room temperature is its reaction with lithium, forming lithium nitride.



React with oxygen in sparking

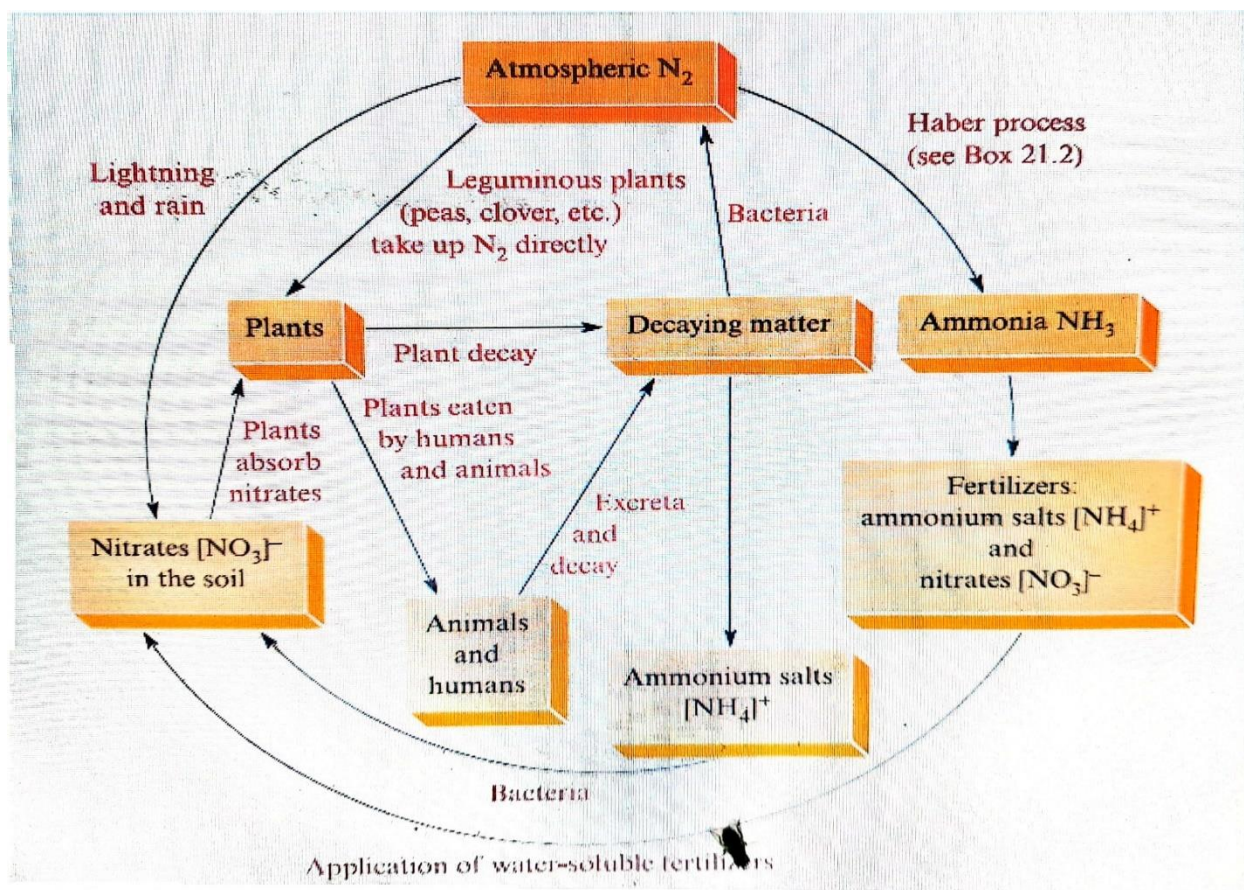


The source of nitrogen is the atmospheric air, and the source of hydrogen is either water or methane.



Nitrogen cycle:

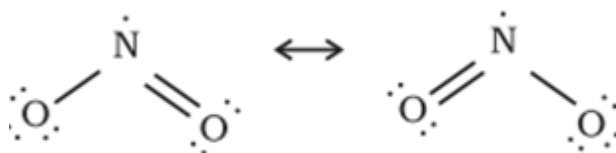
Most organisms cannot use dinitrogen from the atmosphere as their source of the element. However, certain soil bacteria, as well as bacteria that live in nodules on the roots of beans, clover, and similar plants, can “fix” nitrogen; that is, they convert dinitrogen to ammonium and nitrate compounds. Plants use these simple nitrogen compounds to make proteins and other complex nitrogen compounds. Animals eat these plants, and other animals eat those animals. Finally, bacteria in decaying organic matter convert the nitrogen compounds back to dinitrogen. In this way, nitrogen in our environment continually cycles from dinitrogen to living organisms and back. Figure below depicts this *nitrogen cycle*



Nitrogen Oxides :

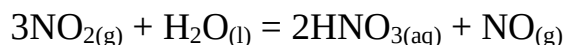
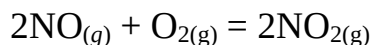
Nitrogen forms many oxides; N_2O , NO , N_2O_3 , NO_2 and N_2O_5 (oxidation numbers ranging from +1 to +5, respectively).

- **Nitrogen dioxide NO_2** Red-brown (a reddish brown gas), Irritating Very toxic

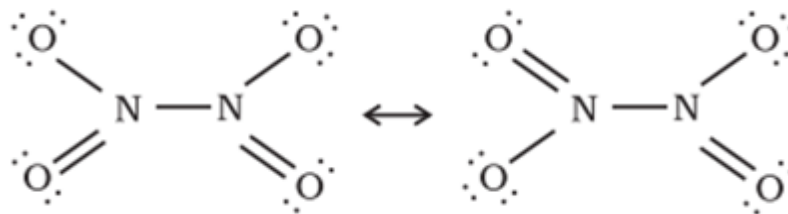


ioxide, NO_2 (a red

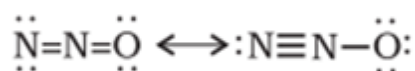
dish brown



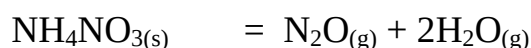
- **Dinitrogen tetroxide N_2O_4** : (a colorless gas).



- **Dinitrogen monoxide**, commonly known as **nitrous oxide**, N_2O , is a colorless gas with a sweet odor.

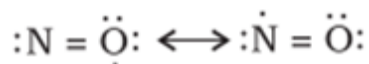


It can be prepared by careful heating of molten ammonium nitrate. (If heated strongly, it explodes.)

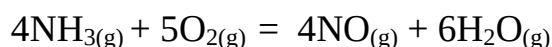


Nitrogen monoxide (NO , also called nitric oxide) Nitrous oxide, or laughing gas, is used as a dental anesthetic. It is also useful as a propellant in whipped-cream dispensers. The gas dissolves in cream under pressure. When the cream is dispensed, the gas bubbles out, forming a foam.

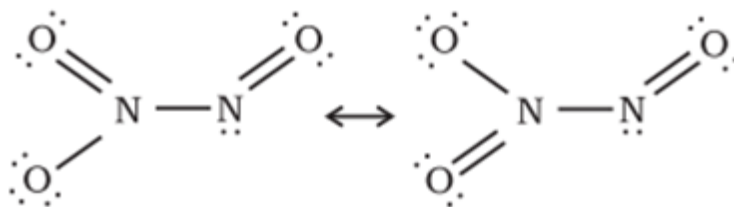
Also plays an important role in biology. Being a small molecule, NO can easily diffuse through cell walls. It behaves as a messenger molecule in biological systems and is active in mammalian functions such as the regulation of blood pressure, muscle relaxation and neurotransmission.



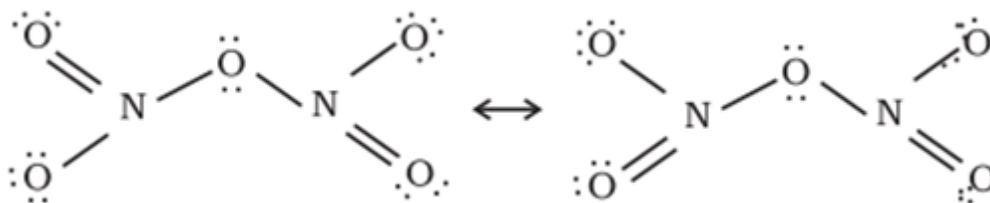
large amounts are prepared from ammonia as the first step in the commercial preparation of nitric acid. The ammonia is oxidized in the presence of a platinum catalyst.



- **Dinitrogen trioxide :**



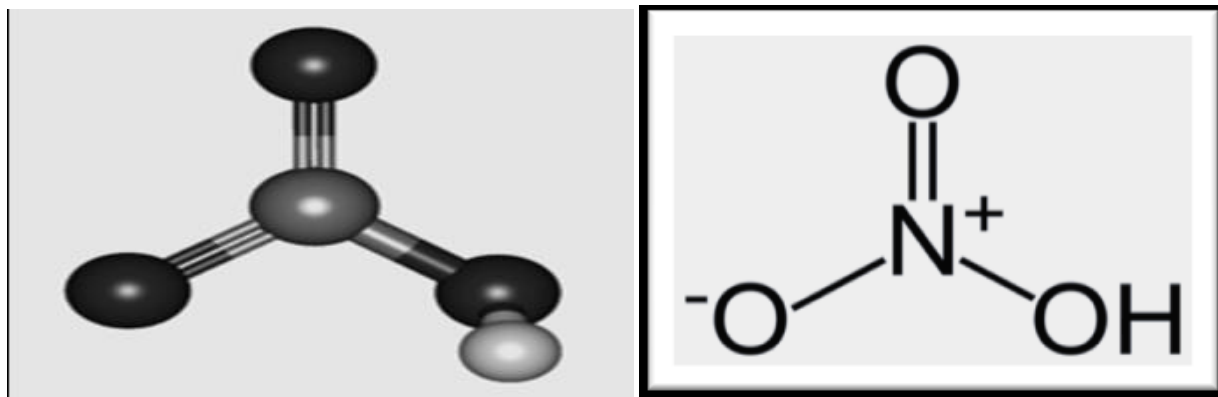
- **Dinitrogen pentoxide N_2O_5 :**



Oxoacid (Oxygenic acids) of Nitrogen

- **Nitric acid: HNO_3**

Its Geometry:



Nitric acid is considered the most important nitrogenous acid. It is colorless liquid (but on exposure to light it turns slightly brown because of slight decomposition into NO_2 and O_2) and boils at 84.1° and freezes into solid crystals at -41° . The pure acid ionizes spontaneously.

Concentrated nitric acid is a strong oxidizing agent for most elements, and its oxidation strength depends on its concentration, as it is noted that the dilute acid is unable to oxidize when diluted to 2M.

it cannot oxidize gold and platinum, so a solution of aqua Regia (consisting of three parts of concentrated hydrochloric acid and one part of concentrated nitric acid).



Where it consists of NOCl and Cl₂, which is attributed to this strong reaction in the formation of complexes with gold and platinum.

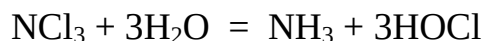
Nitric acid, HNO₃ is the third most important industrial acid (after sulfuric and phosphoric acids). It is used to prepare explosives, nylon, and polyurethane plastics.

- **Nitrous acid HNO₂**

It is also weak acid, it has two states, Cis and Trans, and the form of Trans is more stable than Cis Nitrous acid is unstable except in dilute solution.

- **Hyponitrous acid H₂N₂O₂** a weak acid, and its ions are called hyponitrous (N₂O₂)²⁻ and they are in the form of trans.

nitrogen halides are least stable. All the trihalides with the exception of NF₃ hydrolyse in aqueous solution.



Hydrazine N₂H₄ : Blowing agent for foamed plastics Water treatment.

In 1867, the Swedish chemist Alfred Nobel discovered that nitroglycerin behaved better when absorbed on diatomaceous earth, a crumbly rock, giving an explosive mixture that Nobel called *dynamite*. Part of the wealth he made from his explosive factories was left in trust to establish the Nobel Prizes. Although explosives are often associated in people's minds with war, they do have many peacetime uses, including road building, mining, and demolition



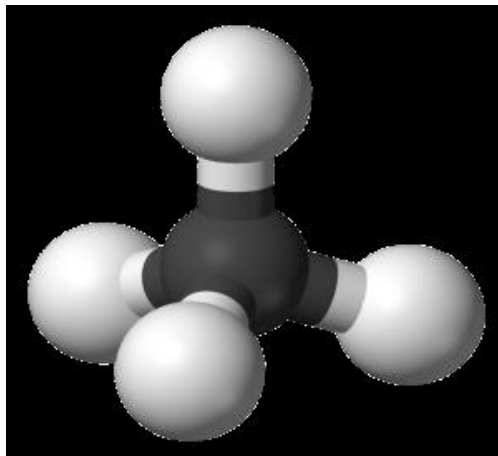
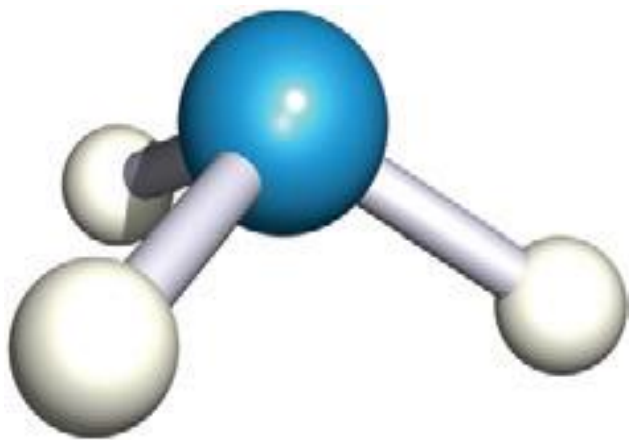
Ammonia

Ammonia, NH_3 is the most important commercial compound of nitrogen. It is a colourless gas (bp 239 K) with a characteristic smell and is corrosive and an irritant or pungent odor, Penetrating Toxic. boils at -33.5 degrees Celsius. Ammonia is very soluble in water and a concentrated solution (density 0.88 g cm^{-3}) is commercially available. The high solubility is due to the extensive hydrogen bonding that occurs between H_2O and NH_3 molecules.

Ammonia is easily liquefied, and the liquid is used as a nitrogen fertilizer. Ammonium salts, such as the sulfate and nitrate, are also sold as fertilizers. Large quantities of ammonia are converted to urea NH_2CONH_2 which is used as a fertilizer, as a livestock feed supplement, and in the manufacture of urea–formaldehyde plastics.

Ammonia is also the starting compound for the preparation of most other nitrogen compounds.

Ammonia molecule is a pyramidal structure in which the hydrogen atoms are located at the corners of an equilateral triangle and the nitrogen atom is at the top of the pyramid. This shape is called the trigonal pyramid



Preparation and reaction of Ammonia:

- As for industrially, it is prepared according to the Haber-Bosch method (Haber Bosch) through direct interaction between hydrogen and nitrogen in the presence of a catalyst at a high pressure of 300 atmospheres and a temperature ranging between 400-500 degrees Celsius. The source of nitrogen is the atmospheric air, and the source of hydrogen is either water or methane.



- the industrial preparation of ammonia from natural gas, steam, and air. Small amounts can be prepared in the laboratory by heating a solution of an ammonium salt with a strong base, such as NaOH or Ca(OH)₂



- Magnesium nitride reacts with water to give magnesium hydroxide and ammonia.



- Combustion in air takes place according to equation

